
A Transfer Learning Approach to Classifying the Intel Image Dataset

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Abstract

This experiment tests whether a convolutional backbone trained with self-supervised rotation prediction transfers to natural-scene classification better than the same architecture trained from scratch. A ResNet-18 was pretrained on the unlabeled split of STL10 to predict rotations of 0° , 90° , 180° , and 270° , then transferred to the six-class Intel Image Classification dataset. Five conditions are compared: S0 (supervised training from scratch), I-Frozen (frozen ImageNet-pretrained backbone), I-FT (full fine-tuning of the ImageNet-pretrained model), SSL-Frozen (frozen STL-10 rotation-pretrained backbone), and SSL-FT (full fine-tuning of the STL-10 rotation-pretrained model). Rotation model validation accuracy reached 89.89%. On Intel, if the model is trained from scratch (S0), it reaches 87.32% validation accuracy and 86.83% test accuracy. If the backbone is ImageNet-pretrained and frozen (I-Frozen), it reaches 91.55% and 90.63% in 8 epochs. If that ImageNet backbone is fully fine-tuned (I-FT), it reaches 94.06% and 92.87% in 10 epochs. If the backbone is the STL10 rotation model and frozen (SSL-Frozen), it reaches 67.09% and 66.83% in 40 epochs. If that rotation backbone is fully fine-tuned (SSL-FT), it reaches 92.64% and 91.80% in 15 epochs. SSL-FT improves on S0 by 5.32 validation percentage points and 4.97 test percentage points, and it remains 1.42 and 1.07 percentage points below I-FT. SSL-Frozen stays 20.00 test percentage points below S0, so the rotation features are a strong initialization and a weak linear representation for this scene task.

1 Methods

1.1 Datasets and splits

Self-supervised pretraining uses the unlabeled split of STL10 that contains 100,000 images. Class labels are ignored. A random 10% of the source images, selected with seed 42, is held out for rotation validation (10,000 images). The remaining 90,000 images are used for rotation training. After rotation, there are in total 360,000 training images and 40,000 validation images. No ImageNet weights and no augmentations other than the four rotations are used at this stage.

Supervised transfer uses the Intel Image Classification dataset. The six classes are buildings, forest, glacier, mountain, sea, and street. The training folder contains 14,034 images: buildings 2,191, forest 2,271, glacier 2,404, mountain 2,512, sea 2,274, and street 2,382. So the training folder is nearly balanced. An 85%/15% split of that folder, again with seed 42, produces 11,928 training images and 2,106 validation images, and 3,000 images test images.

1.2 Model, losses, and optimization

Every condition uses torchvision ResNet-18. The final fully connected layer is replaced by an identity map, so the backbone returns a 512-dimensional vector.

Rotation pretraining uses a 4-way linear head and `nn.CrossEntropyLoss()`. The backbone starts from random weights. Training uses SGD with learning rate 0.05, momentum 0.9, and weight decay 5×10^{-4} , plus cosine annealing over 40 epochs. The batch size is 128. Early stopping monitors rotation validation accuracy with patience 8 and restores the best weights. In this run the model trained for all 40 epochs. After training, only the backbone state dictionary is saved, as `ssl_backbone.pth`.

Intel classification uses a 6-way linear head and the same cross-entropy loss. Training images use a random resized crop to 224×224 with scale (0.7, 1.0) and a random horizontal flip. Validation and test images use a deterministic resize to 256 followed by a center crop to 224. The same ImageNet normalization is applied. The batch size is 64. The optimizer is Adam with weight decay 10^{-4} . From-scratch and frozen runs use learning rate 10^{-3} . Full fine-tuning runs use learning rate 10^{-4} . Each condition may train for up to 40 epochs. Early stopping monitors Intel validation accuracy with patience 5 and restores the best checkpoint before the test evaluation.

The five conditions share this architecture and this data pipeline. S0 trains a randomly initialized network end to end. I-Frozen and I-FT load `ResNet18_Weights.IMAGENET1K_V1`. I-Frozen trains only the classifier. I-FT trains the full network. SSL-Frozen and SSL-FT load `ssl_backbone.pth`. SSL-Frozen trains only the classifier. SSL-FT trains the full network.

1.3 Compute

Both notebooks ran in Google Colab on an NVIDIA T4 GPU with CUDA. The random seed is 42 for each training run.

2 Results

2.1 Rotation pretraining

The rotation model reaches 89.89% validation accuracy (Figure 1). Training accuracy approaches 100% while validation accuracy levels off near 90%. Training loss falls close to zero. Validation loss stays in a higher band and does not follow the training loss down, so the pretext fit is strong and somewhat overfit.

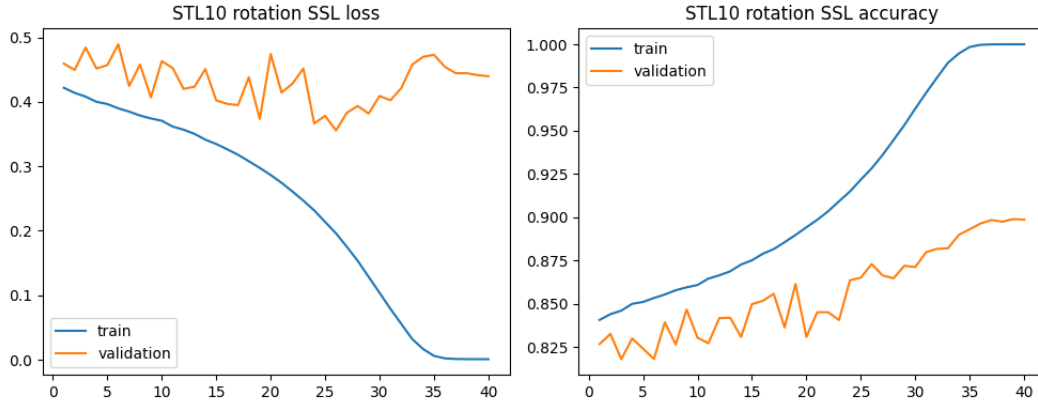


Figure 1: STL10 rotation-prediction loss and accuracy.

2.2 Intel transfer

Table 1 reports the best validation accuracy and the test accuracy of that checkpoint. SSL-FT reaches 92.64% validation accuracy and 91.80% test accuracy after 15 epochs. S0 reaches 87.32% and 86.83% after 20 epochs. The SSL-FT gains are 5.32 percentage points on validation and 4.97 percentage points on test, both above the 2-point improvement criterion.

I-FT is the strongest model, at 94.06% validation and 92.87% test accuracy. SSL-FT trails it by 1.42 percentage points on validation and 1.07 percentage points on test. I-Frozen, which trains

only a classifier on ImageNet features, already reaches 90.63% test accuracy in 8 epochs. SSL-Frozen reaches 67.09% validation and 66.83% test accuracy and uses the full 40-epoch budget while validation accuracy is still rising slowly.

Table 1: Intel Image Classification results.

Condition	Initialization	Frozen?	Epochs	Val (%)	Test (%)
S0	Random	No	20	87.32	86.83
I-Frozen	ImageNet	Yes	8	91.55	90.63
I-FT	ImageNet	No	10	94.06	92.87
SSL-Frozen	STL10 rotation SSL	Yes	40	67.09	66.83
SSL-FT	STL10 rotation SSL	No	15	92.64	91.80

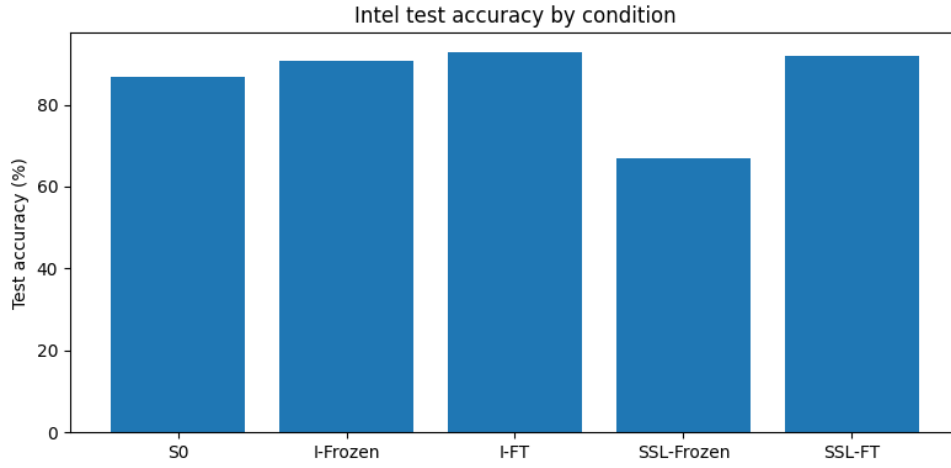


Figure 2: Final Intel test accuracy for the five conditions.

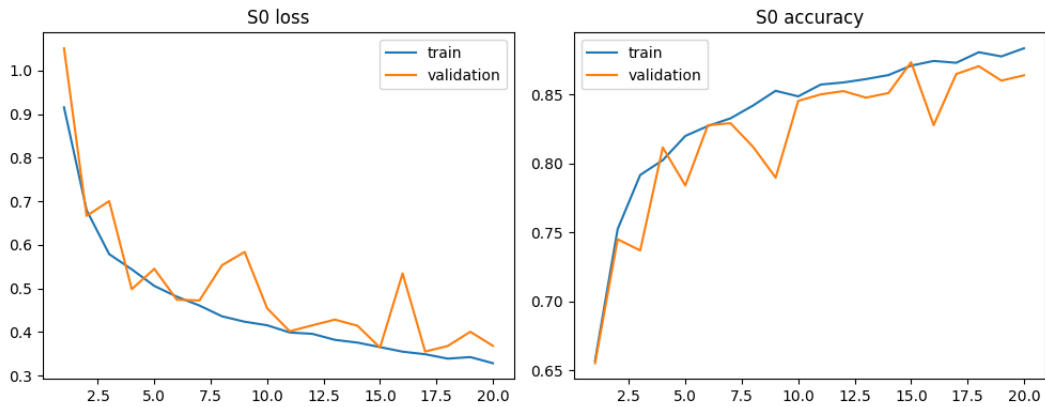


Figure 3: S0 training and validation loss and accuracy.

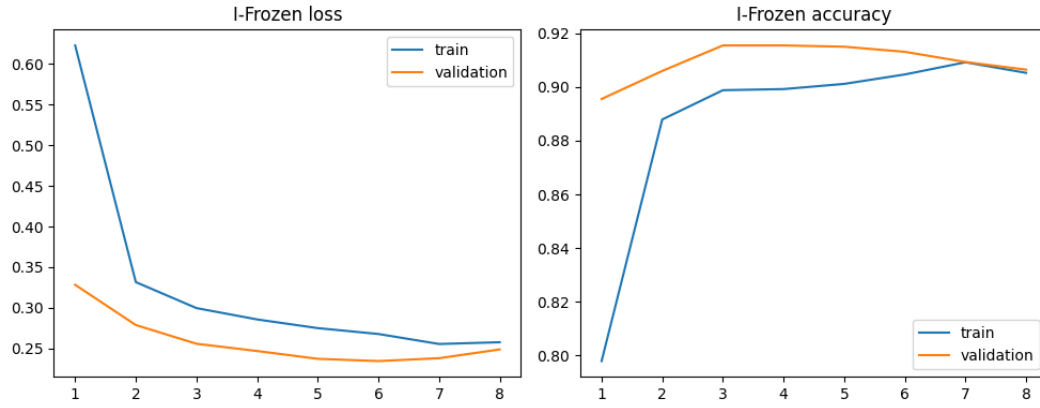


Figure 4: I-Frozen training and validation loss and accuracy.

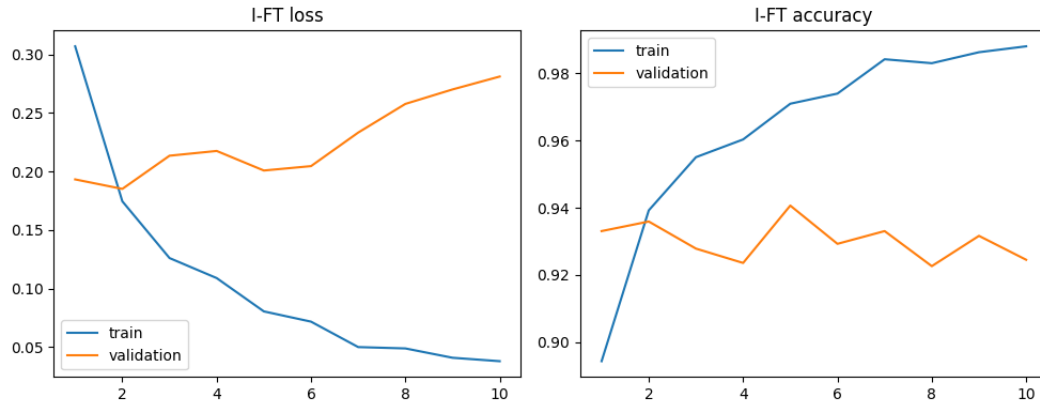


Figure 5: I-FT training and validation loss and accuracy.

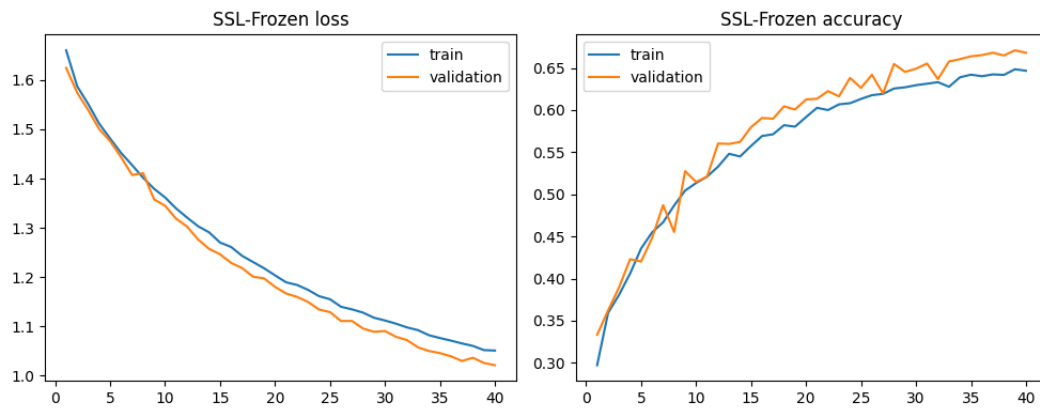


Figure 6: SSL-Frozen training and validation loss and accuracy.

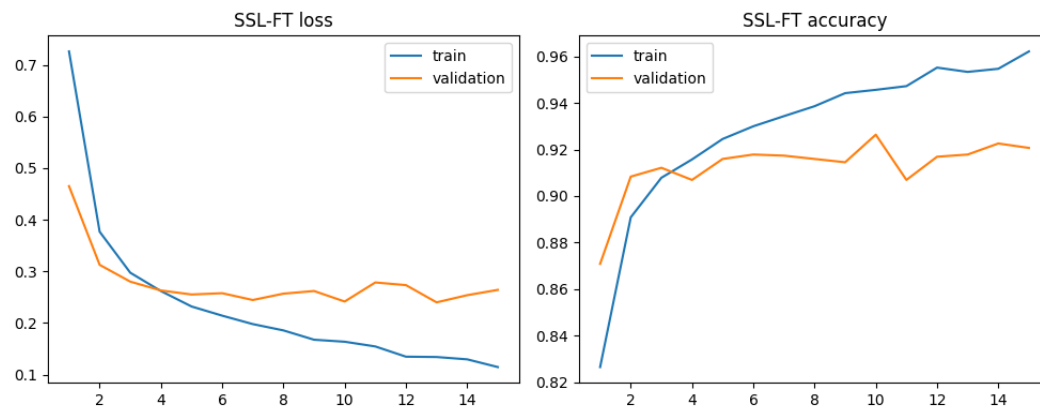


Figure 7: SSL-FT training and validation loss and accuracy.

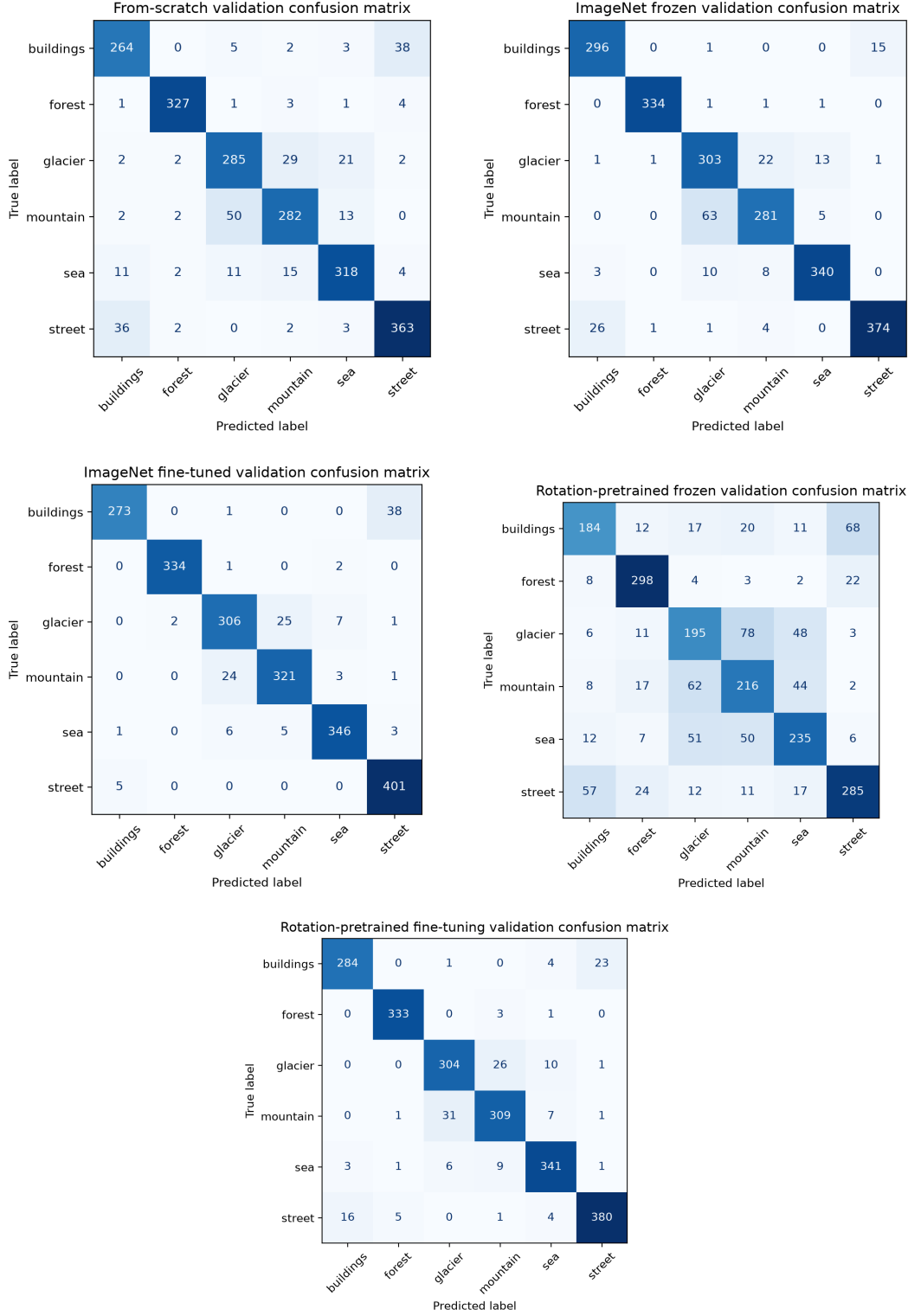


Figure 8: Validation confusion matrices for every condition. Top: S0 and I-Frozen. Middle: I-FT and SSL-Frozen. Bottom: SSL-FT.

3 Discussion

SSL-FT outperforms S0 on both validation and test accuracy. The test improvement is 4.97 percentage points (91.80% versus 86.83%), and the validation improvement is 5.32 percentage points (92.64% versus 87.32%). Both gaps clear the 2-point requirement.

SSL-Frozen does not perform well. Its 66.83% test accuracy is about 20 percentage points below S0 and about 24 percentage points below I-Frozen. It says that the rotation-pretrained backbone is a weak linear representation for these six scene classes.

Rotation SSL gets close to ImageNet pretraining once both backbones are fine-tuned. With full fine-tuning, rotation SSL came within 1.07 test percentage points of ImageNet pretraining: SSL-FT reached 91.80% and I-FT reached 92.87%

The classes confused most often are glacier with mountain, and buildings with street. Those pairs share layout: ice and rock slopes, and built facades along a road.

Rotation prediction can help natural-scene classification because scenes have a stable orientation.

The same objective remains weaker than ImageNet pretraining because it never asks the model to separate object or scene categories. ImageNet pretraining is supervised classification over many labeled object classes, so the frozen features are already organized by meaning.

The next change should make the backbone less tied to the four rotation logits, because SSL-Frozen stalled at 66.83% while SSL-FT reached 91.80%. So I would try to train a small projection head for the rotation loss and discard it before Intel training, so the 512-dimensional features are not specialized only for predicting 0° , 90° , 180° , and 270° .

4 Reproducibility

The notebooks are `ssl_pretrain_new.ipynb` and `intel_transfer_new.ipynb`. They were executed in Google Colab on an NVIDIA T4. Python, PyTorch, and TorchVision were the Colab CUDA runtime defaults for that session.

Shared controls are seed 42. Rotation training uses batch size 128, SGD with learning rate 0.05, momentum 0.9, weight decay 5×10^{-4} , cosine annealing over a 40-epoch horizon, and validation-accuracy patience 8. Intel training uses batch size 64, Adam, weight decay 10^{-4} , a maximum of 40 epochs, and validation-accuracy patience 5. Learning rates are 10^{-3} for S0, I-Frozen, and SSL-Frozen, and 10^{-4} for I-FT and SSL-FT.

The rotation backbone is saved as `ssl_backbone.pth` under `MyDrive/ssl_pretrain/` on Google Drive. Intel checkpoints are `s0.pth`, `i_frozen.pth`, `i_ft.pth`, `ssl_frozen.pth`, and `ssl_ft.pth`.

To rerun, execute `ssl_pretrain_new.ipynb` from top to bottom on a GPU so that it downloads STL10 and writes the backbone checkpoint. Then execute `intel_transfer_new.ipynb`, with the Intel zip available at the Drive path set in that notebook and with `SSL_BACKBONE_PATH` pointing at `ssl_backbone.pth`. The second notebook trains S0, I-Frozen, I-FT, SSL-Frozen, and SSL-FT in that order and writes the curves, confusion matrices, and summary table.